

Firewall and IDPS Documentation

PrimeCore Minerals | Secure Multi Office Network

PC-12 | Version 0.1 | 23 September 2026
Status: Partly evidenced firewall and proposed IDPS
Owner: IT Administrator | Approver: Project lead | Approval date: Pending

Describe the evidenced HQ firewall control and central alerting path, then specify a proposed intrusion detection and prevention extension. A firewall block count is distinct from a deployed signature-based IDPS.

Architecture and observed control

HQ-FW routes between the recorded VLANs. R06 on EMPLOYEE_WIFI_VLAN blocks the management network 192.168.99.0/24 with logging, tracker 1787829015. T01 confirms only the test from 192.168.11.100 to firewall address 192.168.99.1 using IPv4 ICMP. Capture the current full rule export before claiming broader coverage. [S1-S3]

Flow	Service and action	Status
VLAN 11 employee to management network	R06 block and log	Tested ICMP path only
HQ-FW source 192.168.90.1 to HQ-MONITOR .60	UDP 514 logging	Received records evidenced
HQ-MONITOR to mail submission service	TCP 587; STARTTLS	Delivery tested; final rule capture outstanding
Approved admin endpoint to management	Only selected management ports	Proposed; source and ports to register
Visitor network to internal ranges	Deny internal access; allow approved Internet services	Proposed; test required
Other inter-VLAN traffic	Explicit business allow then deny	Full ruleset review required

Rule implementation method

For ordinary interface rules, evaluate new connections on ingress and keep specific controls ahead of broad allows. Review floating rules, interface groups, NAT and existing states before drawing conclusions from order. Use aliases with documented members, owner and purpose. Record source, destination, protocol, port, action, logging and reason for every rule. Revalidate saved states carefully after an authorised rule change. [N1]

IDPS design and platform comparison

Option	Fit and tradeoff	Decision
Existing pfSense virtual firewall	Reuses demonstrated lab; shares virtual host failure and capacity risks	Retain for lab validation
Dedicated firewall appliance	Independent hardware; requires measured throughput, support and port requirements	Evaluate for production; no model selected
Managed commercial platform	Integrated support and policy tooling; licence and migration assessment required	Alternative for procurement review
Suricata or Snort package on pfSense	Adds IDS/IPS capability; requires compatibility, interface and performance testing	Proposed detection-only pilot

Netgate identifies Snort and Suricata as add-on IDS/IPS options. No installed package, active signature feed or prevention test is evidenced. Start a selected package in detection-only mode on a chosen inspection path, tune against normal traffic and introduce blocking only after false-positive and performance checks. [N2]

1. Record installed pfSense version, supported package version, interface placement, VLAN visibility and available CPU, memory and NIC support. Back up configuration and maintain console access.
2. Define protected networks and explicit exclusions. Choose and document a permitted signature feed, update schedule and rule categories. Record feed version with test results.
3. Generate harmless, labelled test traffic on authorised lab endpoints. Confirm the expected signature, source, destination and central alert, then review legitimate application traffic for false positives.
4. If prevention is approved, enable only validated rules, repeat application and performance tests, and prove an allowed control flow still works. Roll back to detection-only if false blocks or resource pressure occur.

Acceptance and remaining work

FW-01: repeat R06 ICMP correlation. FW-02: test employee HTTPS/SSH to a designated management host and an allowed admin path. IDS-01: show a signature alert. IPS-01: show the selected prohibited flow blocked and a legitimate control flow allowed. Measure throughput and resource use against agreed requirements. Capture rule exports, signature ID, timestamps, packet or log evidence and reviewer. Remaining R06 tests include exact-ten boundary, multiple-source isolation and independent collector health.

References and linked artefacts

[S1] HQ_Network_Addressing_Artefact.md, project deliverables, evidence dated 13 September 2026.

[S2] Security_Alerting_Implementation_Artefact.md, project deliverables, 13 September 2026. Credits Jared Williams for the original monitoring foundation and Adam Donovan for subsequent remediation, alerting and validation.

[S3] Security_Test_Results.md, project deliverables, 13 September 2026, tests T01 to T05 and their evidence limitations.

[N1] Netgate. Rule Methodology. <https://docs.netgate.com/pfsense/en/latest/firewall/rule-methodology.html> (accessed 23 September 2026).

[N2] Netgate. IDS / IPS. <https://docs.netgate.com/pfsense/en/latest/packages/snort/index.html> (accessed 23 September 2026).

Screenshot evidence appendix

These figures record selected laboratory observations. They support the tested functions described in this artefact. Proposed policies, untested controls and organisational approval remain separate from the pictured results.

Figure E1 Employee WiFi firewall rules

Firewall / Rules / EMPLOYEE_WIFI_VLAN

Floating WireGuard WAN LAN EMPLOYEE_VLAN VISITOR_VLAN EMPLOYEE_WIFI_VLAN PRINTER_VLAN VIDEO_VLAN
CCTV_VLAN IOT_VLAN SERVER_VLAN BACKUP_VLAN MANAGEMENT_VLAN

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗ 0/0 B	IPv4 *	EMPLOYEE_WIFI_VLAN subnets	*	MANAGEMENT_NETWORK	*	*	none		R06 - Block Employee WiFi Access to Management	
☐	✓ 0/0 B	IPv4 ICMP 80XX	EMPLOYEE_WIFI_VLAN subnets	*	*	*	*	none		TEMP - Employee wifi connection	
☐	✓ 0/16.00 MiB	IPv4 TCP	EMPLOYEE_WIFI_VLAN subnets	*	*	WEB_PORTS	*	none		R03a - Allow Employee WiFi Web Access	
☐	✓ 0/0 B	IPv4 TCP	EMPLOYEE_WIFI_VLAN subnets	*	PRINTER_NETWORK	PRINT_PORTS	*	none		R09b - Allow Employee WiFi for Printers	
☐	✓ 0/895 KiB	IPv4 TCP/UDP	EMPLOYEE_WIFI_VLAN subnets	*	*	DNS_PORT	*	none		R03b - Allow Employee WiFi DNS	
☐	✓ 1/2.34 MiB	IPv4 TCP	EMPLOYEE_WIFI	*	AD_SERVER	AD_TCP_PORTS	*	none			
☐	✓ 0/0 B	IPv4 UDP	EMPLOYEE_WIFI	*	AD_SERVER	AD_UDP_PORTS	*	none			

Add Add Delete Toggle Copy Save Separator

The displayed rule order includes R06 blocking management access. The controlled validation below covers IPv4 ICMP to 192.168.99.1, rather than every management service.

Source evidence: evidence/HQ_EMPLOYEE_WIFI_Rules.png

Screenshot evidence continued

Figure E2 Logged management block

Firewall / Rules / Edit

Edit Firewall Rule

Action

Block

Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled

☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

EMPLOYEE_WIFI_VLAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

Any

Choose which IP protocol this rule should match.

Source

Source

☐ Invert match

EMPLOYEE_WIFI_VLAN subnets

Source Address

/

Destination

Destination

☐ Invert match

Address or Alias

MANAGEMENT_NETWORK

/

Extra Options

Log

☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

Description

R06 - Block Employee WIFI Access to Management

A description may be entered here for administrative reference. A maximum of 57 characters will be used in the ruleset label and displayed in the firewall log.

Advanced Options

☒ Display Advanced

Rule Information

Tracking ID

1787829015

Created

8/27/26 11:10:15 by admin@10.10.10.102 (Local Database)

Updated

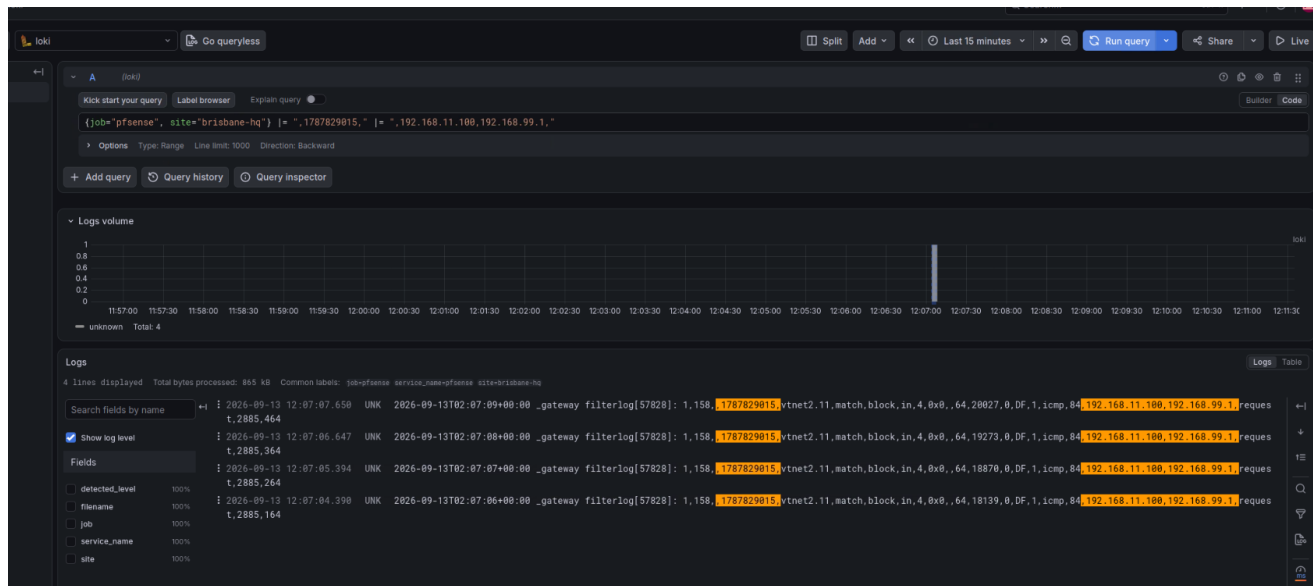
8/27/26 11:10:15 by admin@10.10.10.102 (Local Database)

The rule configuration shows logging enabled and tracker 1787829015, supporting correlation with centrally collected events.

Source evidence: evidence/HQ_R06_Logging_Tracking_ID.png

Screenshot evidence continued

Figure E3 Correlated firewall records



Grafana retrieves four matching R06 records for source 192.168.11.100 and destination 192.168.99.1. This establishes the demonstrated collection and retrieval path.

Source evidence: evidence/T01_Grafana_R06_Four_Blocks.png